COVID-19 RAG Pipeline

Technical How-To Guide · Databricks Free Edition · MLflow Tracked

1. Pipeline Overview

We built a five-notebook Databricks pipeline to answer COVID-19 questions from uploaded PDF documents. PDFs are ingested, chunked, embedded, and indexed — then served through an agentic RAG chain backed by Meta Llama 3.3 70B. All data persists in Unity Catalog (main.covid_rag); all experiments are tracked in MLflow at /Shared/covid_rag_experiments.

Notebook	Stage	What It Does
01_setup	Environment	Installs all dependencies: langchain, sentence-transformers, faiss-cpu, mlflow, pypdf, pandas
02_ingest_pdfs	Ingestion	Reads every PDF page-by-page with PyPDF; drops pages < 50 chars; saves to main.covid_rag.covid_pages Delta table
03_chunk_embed_index	Chunk & Embed	Splits pages into 800-char chunks (150 overlap); encodes with all-MiniLM-L6-v2 (384-dim); builds FAISS IndexFlatL2; saves index + metadata to Volume
04_rag_chain	Query & Generate	Implements 3-route agentic logic (greeting / internal / Wikipedia fallback); calls Llama 3.3 70B; keeps 6-turn conversation memory; auto-logs to MLflow
05_evaluation	Evaluation	Runs 12 test questions; logs route accuracy, latency, and answer artifacts to MLflow; generates manual scoring template

2. How We Built It

Data Ingestion

PDFs were stored in /Volumes/main/covid_rag/covid_docs/. We listed files with dbutils.fs.ls(), extracted text page-by-page using PyPDF's PdfReader, and filtered out pages under 50 characters. The result was saved to the covid_pages Delta table via spark.createDataFrame().write.saveAsTable().

Chunking, Embedding & Indexing

Pages were split using LangChain's RecursiveCharacterTextSplitter (chunk_size=800, chunk_overlap=150), which breaks on paragraph boundaries first to keep chunks semantically intact. Each chunk got a unique ID (filename + page + index). We then batch-encoded all chunks with all-MiniLM-L6-v2 to produce 384-dim vectors, stored in covid_chunk_embeddings. A FAISS IndexFlatL2 (exact L2 search) was built over all vectors and saved to the Volume as faiss_index.bin, alongside a pickled chunk DataFrame for mapping results back to source text.

Agentic Routing & Generation

Every query enters answer_query() and is routed through three checks:

- **Greeting** → Keyword-matched (hello, hi, how are you, etc.) and returns a static response instantly — no retrieval needed.
- **Internal** → Query is embedded and top-4 chunks are fetched from FAISS. A quality gate (≥ 2 chunks, ≥ 250 chars, avg L2 distance ≤ 1.2, keyword overlap) decides if context is good enough. If yes, the internal prompt template is used with Llama 3.3 70B (temperature=0.1, max_tokens=400).
- **External** → If the gate fails, fetch_wikipedia_summary() pulls a Wikipedia summary and a YouTube link is appended. The external prompt handles general fallback questions.

Conversation memory retains the last 6 turns (formatted as User:/Assistant:) and is injected into every prompt, enabling follow-up questions. mlflow.langchain.autolog() traces every LLM call automatically.

Evaluation

We tested 12 questions: 9 routed internally (symptoms, transmission, prevention, vaccination, long COVID, treatment, public health guidance) and 3 routed externally (travel restrictions, live case counts, airline policies). Each call was timed and the route was compared against the expected one. Four metrics — route_accuracy,

avg_total_latency_sec, internal_route_rate, external_route_rate — were logged to MLflow with a full per-question artifact table.

3. Reading the MLflow Results

Go to Experiments → /Shared/covid_rag_experiments → open run covid_rag_evaluation.

Metric	Good Range	What It Tells You
route_accuracy	≥ 0.80	% of questions correctly routed. Below 0.80 means the L2 distance or keyword-overlap thresholds need adjustment.
avg_total_latency_sec	< 5 sec	Mean end-to-end response time. Spikes come from the Wikipedia API or LLM — not from FAISS, which is sub-millisecond.
internal_route_rate	≈ 0.75	Share of questions answered from PDFs. Theoretical ceiling is 0.75 for our 9-internal / 3-external split.
external_route_rate	≈ 0.25	Share falling back to Wikipedia. Higher than 0.25 signals the PDFs lack coverage for those topics.

The Artifacts tab contains evaluation_results.json with one row per question (route, latency, sources, answer). The Traces tab shows the full prompt sent to Llama for each call — useful for debugging individual answer quality.

4. What We Learned

- **Routing quality matters more than retrieval quality.** The quality gate prevented the LLM from hallucinating answers from loosely related chunks — routing to Wikipedia instead was a better user experience than a confident wrong answer.
- **Chunk size directly affects answer completeness.** Chunks that were too small lost context mid-sentence; chunks too large diluted relevance scores. The 800/150 split hit the right balance for our COVID documents.
- **MLflow tracing is essential for debugging.** Without autolog traces, it was nearly impossible to tell whether a poor answer came from bad retrieval or bad generation. The traces made this immediately visible.
- **FAISS L2 distance is not a perfect relevance signal.** Some chunks with low L2 distances were syntactically similar but semantically irrelevant, which is why the keyword-overlap check in the quality gate was necessary as a secondary filter.
- **Conversation memory needs a hard cap.** Without the 6-turn limit, the injected history grew long enough to push the actual context out of the model's attention window, degrading answer quality on follow-ups.